

MODULE 17 · DOMAIN 2 - DESIGN FOR NEW SOLUTIONS

Choosing a Database

Relational, key-value, or purpose-built. Read the access pattern, not the data.

DOMAIN 2 of 4 EXAM WEIGHT 29% MODULE 17 of 44

The question is not what the data looks like.
It is **how it is queried**, and whether the
pattern is known in advance.

Relational or DynamoDB?

AURORA OR RDS

Unknown queries, joins, SQL

- Ad hoc queries and reporting
- Joins and multi-table transactions
- Existing SQL applications
- Scales up, and read replicas out

DYNAMODB

Known access patterns, huge scale

- Single-digit ms at any volume
- Access patterns fixed at design time
- No servers, no version upgrades
- Scales horizontally without limit

THE TIEBREAKER

Unbounded scale with predictable queries means DynamoDB. Anything needing joins or unforeseen queries means relational.

RDS or Aurora?

- ◆ **Aurora separates compute from storage**, with a shared volume across three AZs and six copies. Replicas share it rather than copying.
- ◆ **Replica lag is milliseconds**, against seconds for RDS, and failover is typically under a minute.
- ◆ **RDS still wins** when you need Oracle, SQL Server, MariaDB or Db2. Aurora is MySQL and PostgreSQL compatible only.
- ◆ **RDS Multi-AZ cluster deployments** give two readable standbys, closing part of the gap.

ON THE EXAM

Aurora storage grows automatically to 128 TiB. Engine choice, not performance, is what usually forces RDS.

Aurora at multi-Region scale

- ◆ **Aurora Global Database** replicates to secondary Regions with typical lag under a second, and cross-Region failover in about a minute.
- ◆ **Write forwarding** lets a secondary Region accept writes and forward them to the primary, so applications need no routing logic.
- ◆ It is still **one writer**. Global Database is active-passive, whatever the write forwarding makes it look like.
- ◆ **Aurora Serverless**, formerly Serverless v2, scales by ACU and now down to zero with auto-pause.

ON THE EXAM

Write forwarding hides the round trip; it does not remove it. A write from Sydney still travels to the primary Region.

DynamoDB, and a changed default

- ◆ **On-demand throughput was cut by 50% in Nov 2024.** AWS now calls on-demand the default and recommended mode for most workloads.
- ◆ **Global tables** are multi-active: every Region accepts writes. Replicated write pricing fell by up to 67% at the same time.
- ◆ **DAX** is a DynamoDB-specific cache giving microsecond reads. **Streams** emit item changes for event-driven work.
- ◆ Global secondary indexes give alternative access patterns. Model the table around queries, not entities.

ON THE EXAM

The old advice was provisioned for steady workloads, on-demand for spiky ones. The 2024 price cut reversed the default.

Global tables or Aurora Global Database?

DYNAMODB GLOBAL TABLES

Every Region writes

- Multi-active, no failover needed
- Eventually consistent across Regions
- Strong consistency now optional
- Last writer wins on conflict

AURORA GLOBAL DATABASE

One Region writes

- Active-passive, promote to fail over
- Sub-second replication lag
- Write forwarding hides the routing
- Full SQL and transactions

THE TIEBREAKER

Writes accepted in every Region with no failover means DynamoDB. If the answer must be relational, accept active-passive.

The purpose-built line-up

SERVICE	DATA SHAPE	SIGNAL IN THE SCENARIO
ElastiCache	In-memory cache or store	Reduce database load, session state
Amazon Neptune	Graph, with relationships	Fraud rings, recommendations, social
Amazon DocumentDB	JSON documents, MongoDB API	Migrating an existing MongoDB estate
Amazon Keyspaces	Wide-column, Cassandra API	Existing Cassandra, CQL
Amazon Timestream	Time series	IoT sensors, metrics over time
Amazon Redshift	Columnar warehouse	Analytics over billions of rows

ON THE EXAM

Purpose-built questions name an API or a data shape. Cassandra, MongoDB or a graph in the wording is the whole answer.

ElastiCache, and a renamed engine

- ◆ **Three engines: Valkey, Redis OSS and Memcached.** Valkey is the open-source Redis fork and is API compatible.
- ◆ **Valkey is roughly 20% cheaper node-based and 33% cheaper serverless**, with a zero-downtime in-place upgrade path.
- ◆ **Redis OSS v4 and v5 left standard support on 31 January 2026** and now carry Extended Support charges.
- ◆ Memcached is multi-threaded and simple. Redis and Valkey add replication, persistence and data structures.

ON THE EXAM

Replication, persistence or sorted sets means Redis or Valkey. Pure horizontal cache with no features means Memcached.

Two notes on cost and scope

- ◆ **Database Savings Plans** now cover Aurora, RDS, DynamoDB, ElastiCache for Valkey, DocumentDB, Neptune, Keyspaces and DMS.
- ◆ **Not Redshift.** One year, no upfront, latest-generation instances only. Redis OSS and Memcached are excluded.
- ◆ **Aurora DSQL** gives active-active multi-Region SQL writes with strong consistency. It reached GA in mid-2025.
- ◆ **DSQL is not on the exam's in-scope list.** Know it exists; answer questions with Aurora Global Database.

ON THE EXAM

DSQL genuinely solves what Global Database cannot. The exam does not list it, so it is not the answer there.

Reading the scenario

THE SCENARIO SAYS

Millions of requests, fixed access patterns

Ad hoc reporting queries and joins

Oracle, SQL Server, MariaDB or Db2

Writes accepted in several Regions at once

Relationships between entities are the query

Database CPU spent on repeated identical reads

IT IS POINTING AT

DynamoDB

Aurora or RDS

RDS, not Aurora

DynamoDB global tables

Amazon Neptune

ElastiCache, or DAX

ON THE EXAM

Name the engine, the access pattern and the write topology. Most database questions are decided by one of those three.

Meridian Freight is designing a parcel tracking service. Every lookup is by tracking number and returns one parcel's status history. Volume is expected to reach 400,000 requests per second at peak, growing yearly. Responses must be under 10 milliseconds.

The team's existing skills are in PostgreSQL.

What should the solutions architect recommend?

- ☐ A Aurora PostgreSQL with read replicas, since the team knows PostgreSQL
- ☐ B DynamoDB, with the tracking number as the partition key
- ☐ C Aurora PostgreSQL with an ElastiCache layer in front
- ☐ D Amazon Keyspaces, for wide-column scale

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Aurora with read replicas
Aurora allows fifteen read replicas. 400,000 requests per second and growing will exhaust that ceiling, and the ceiling is fixed.
- B** DynamoDB
One access pattern, always by key, at unbounded scale with a millisecond target. This is precisely what DynamoDB is for.
- C** Aurora plus ElastiCache
The strong near miss. It would work today, and it adds a cache to manage and invalidate, and Aurora remains the ceiling underneath.
- D** Keyspaces
Also scales, and is for teams with existing Cassandra applications and CQL. Nothing here suggests either.

THE SIGNAL

Team skills are real and they are not a requirement. The access pattern and the growth curve are.

A retailer runs a global storefront. Customers in Europe, North America and Asia all place orders, and an order written in any Region must be visible in the others within seconds. Order records are simple documents retrieved by order ID.

The business will not accept a failover step before writes can resume in a Region.

What should the solutions architect recommend?

- ☐ A Aurora Global Database with write forwarding from the secondary Regions
- ☐ B DynamoDB global tables across the three Regions
- ☐ C Aurora in one Region with read replicas in the others
- ☐ D Three independent DynamoDB tables, synchronised nightly

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Global Database with write forwarding
The sharpest distractor. Writes are accepted everywhere and forwarded to the single primary, so losing the primary still needs a promotion.
- B

 DynamoDB global tables
Genuinely multi-active. Every Region owns writes, so there is no primary to lose and no failover step at all.
- C

 Read replicas
Replicas are read-only. Two of the three Regions could not accept an order at all.
- D

 Nightly sync
Seconds was the requirement. Nightly is roughly fifty thousand times too slow, and invites conflicting order IDs.

THE SIGNAL

No failover step is the phrase. Write forwarding removes the routing logic, not the single primary underneath it.

A DynamoDB table serves an application whose traffic is steady at roughly 4,000 read units and 1,500 write units, all day, every day. It has used provisioned capacity with auto scaling since 2022.

A cost review asks whether the capacity mode is still the right choice.

What should the solutions architect advise?

- A Keep provisioned capacity; it is cheaper for steady workloads
- B Compare against on-demand, which AWS now recommends as the default after the 2024 price cut
- C Switch to provisioned capacity with reserved capacity for a further discount
- D Move the table to Aurora, which is cheaper at steady load

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Keep provisioned

The correct answer until November 2024. AWS halved on-demand pricing and now states most provisioned workloads are cheaper on-demand.

C Reserved capacity

Deepens a commitment to a mode that may no longer be the cheaper one. Measure before committing for a year.

B Compare against on-demand

The price cut changed the default. Modelling the actual numbers is the right advice, and on-demand also removes capacity management.

D Move to Aurora

Replaces the database to change a billing mode, and Aurora would not serve this access pattern better.

THE SIGNAL

Since 2022 is the planted detail. The design was correct when it was made and the pricing moved underneath it.

An insurer is migrating a claims system that runs on Oracle with PL/SQL packages, materialised views, and a third-party actuarial extension. It needs Multi-AZ resilience and automated backups, and the business will not fund an application rewrite.

An architect has proposed Aurora PostgreSQL for its performance.

What should the solutions architect advise?

- ☐ A Aurora PostgreSQL, using BabelFish for compatibility
- ☐ B Amazon RDS for Oracle in a Multi-AZ deployment
- ☐ C Oracle on EC2 with a self-managed standby
- ☐ D Amazon DocumentDB, since claims are documents

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

Aurora with Babelfish

Babelfish translates SQL Server T-SQL, not Oracle PL/SQL. It is the wrong compatibility layer for this engine entirely.
- B

RDS for Oracle, Multi-AZ

Keeps the engine, so PL/SQL and the extension keep working, while AWS handles Multi-AZ and backups. No rewrite.
- C

Oracle on EC2

Preserves the engine and hands back patching, backup and standby management that RDS provides.
- D

DocumentDB

Claims being called documents in English does not make them MongoDB documents. This is a full rewrite.

THE SIGNAL

Babelfish is for SQL Server, not Oracle. Knowing which compatibility layer belongs to which engine is the whole question.

A payments platform's Aurora writer is at 85% CPU. Analysis shows 70% of queries are identical reads of a small set of merchant configuration rows that change a few times a day. The team wants to reduce load without changing the schema or sharding.

What should the solutions architect recommend?

- ☐ A Add Aurora read replicas and route reads to them
- ☐ B Add an ElastiCache layer caching the configuration rows, invalidated on change
- ☐ C Scale the writer to a larger instance class
- ☐ D Add DynamoDB Accelerator in front of Aurora

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Read replicas

The strong near miss. It moves the reads off the writer, and you still execute the same identical query millions of times, just elsewhere.

C Bigger writer

Buys headroom for the same wasteful pattern, and instance classes run out. It postpones rather than solves.

B ElastiCache with invalidation

Data read constantly and changed a few times a day is the textbook cache. The query stops executing rather than moving.

D DAX in front of Aurora

DAX is DynamoDB-specific. It cannot sit in front of Aurora at all.

THE SIGNAL

Read often, changed rarely, small dataset. Three properties that together describe a cache and nothing else.

A bank is building fraud detection. The core query is finding chains of relationships between accounts, devices and addresses, up to six hops deep, in real time. The data is highly connected and the useful questions are about paths rather than rows.

An architect proposes recursive SQL on Aurora PostgreSQL.

What should the solutions architect recommend?

- ☐ A Aurora PostgreSQL with recursive common table expressions
- ☐ B Amazon Neptune
- ☐ C DynamoDB with a global secondary index per relationship type
- ☐ D Amazon Redshift, since it handles large joins

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Recursive CTEs on Aurora

Expressible in SQL, and each hop is another self-join. At six hops in real time the query planner will not hold up.

C DynamoDB with GSIs

Each hop becomes a separate query the application stitches together. Traversal logic moves into your code and gets slower with depth.

B Amazon Neptune

A graph database traverses relationships as a first-class operation, so six hops costs roughly six steps rather than six joins.

D Redshift

Columnar and built for aggregation over many rows. It is an analytics warehouse, and this is a real-time traversal.

THE SIGNAL

Paths rather than rows is the phrase. When relationships are the query rather than an implementation detail, that is a graph.

What to carry out of this module

- ◆ **Ask how it is queried, not what it looks like.** Known access patterns at scale means DynamoDB.
- ◆ **Aurora Global Database is active-passive.** Write forwarding hides the routing, not the single primary.
- ◆ **DynamoDB on-demand is now the default** after the 2024 price cut. Older material still says provisioned.
- ◆ Babelfish is for SQL Server, not Oracle. DAX is for DynamoDB only. Both are precise knowledge checks.

ON THE EXAM

Engine support is what usually forces RDS over Aurora. Oracle, SQL Server, MariaDB and Db2 have no Aurora equivalent.